

Multiple Choice (10 marks)

1. The sign of the charge carriers in a doped semiconductor can be deduced by measuring which of the following properties?
 - (a) Magnetic susceptibility
 - (b) Hall coefficient
 - (c) Electrical resistivity
 - (d) Thermal conductivity
2. A single-electron atom has the electron in the $l = 2$ state. The number of allowed values of the quantum number m_l is
 - (a) 5
 - (b) 4
 - (c) 3
 - (d) 2
3. A rod measures 1.00 m in its rest system. How fast must an observer move parallel to the rod to measure its length to be 0.80 m?
 - (a) 0.50 c
 - (b) 0.60 c
 - (c) 0.70 c
 - (d) 0.80 c
4. De Broglie hypothesized that the linear momentum and wavelength of a free massive particle are related by which of the following constants?
 - (a) Planck's constant
 - (b) Boltzmann's constant
 - (c) The Rydberg constant
 - (d) The speed of light
5. A refracting telescope consists of two converging lenses separated by 100 cm. The eyepiece lens has a focal length of 20 cm. The angular magnification of the telescope is
 - (a) 4
 - (b) 5
 - (c) 6
 - (d) 20

True / False (10 marks)

Answer True or False

6. True or False: Fermions have antisymmetric wave functions and obey the Pauli exclusion principle.
7. True or False: A particle moving at $v=0.60\ c$ will travel 288 m before decaying in the lab frame.
8. True or False: The angular magnification of the refracting telescope is 4, given the eye-piece lens has a focal length of 20 cm.
9. True or False: The mean kinetic energy of conduction electrons in metals is high due to their many degrees of freedom compared to atoms.
10. True or False: If the absolute temperature of a blackbody is increased by a factor of 3, the energy radiated per second per unit area increases by a factor of 81.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the total spin quantum number of the electrons in the ground state of neutral nitrogen ($Z = 7$), explaining how you arrive at your conclusion and its significance in atomic structure.
12. Discuss the electron configuration of an atom with filled $n = 1$ and $n = 2$ energy levels, and analyze how this configuration determines the total number of electrons in the atom.

End of Paper